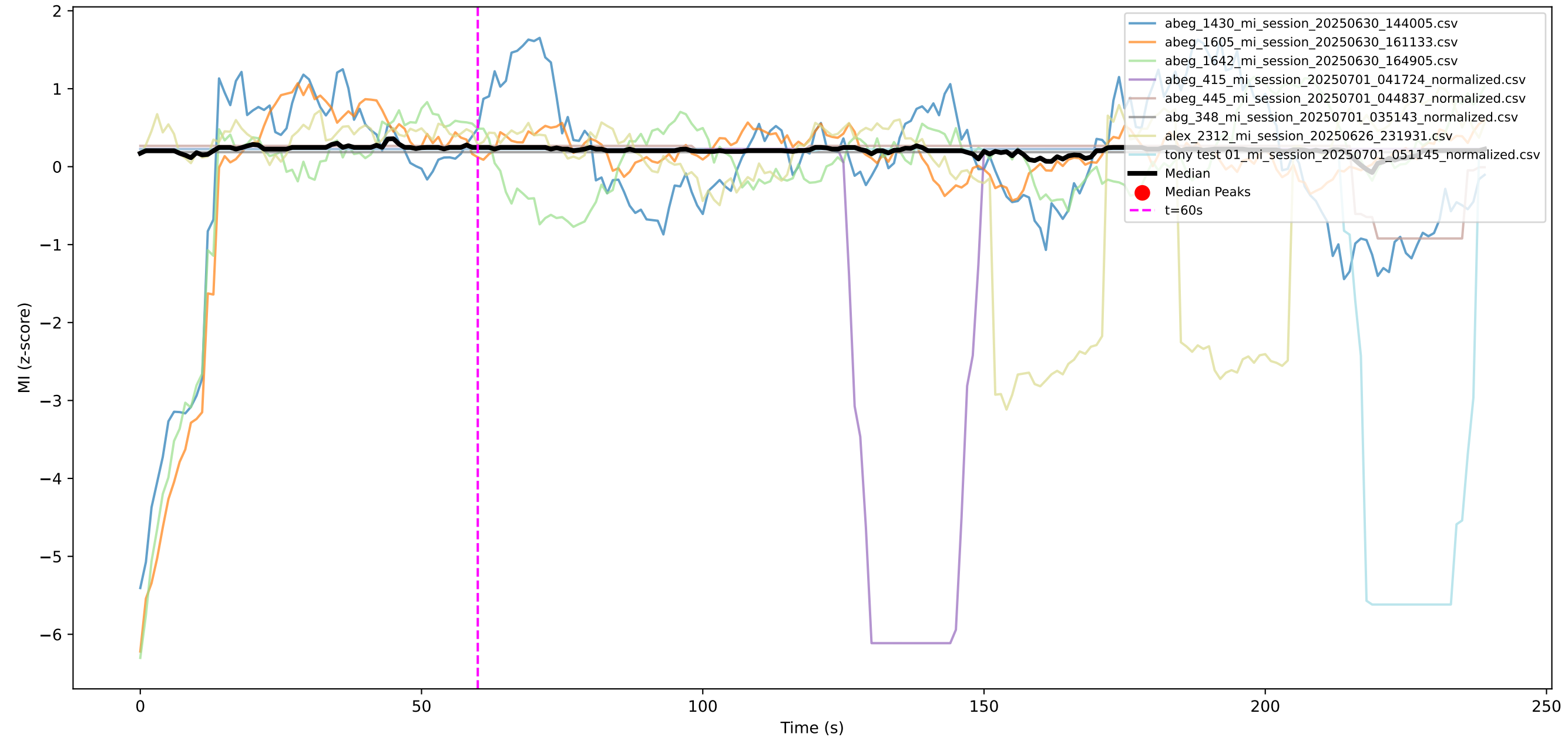
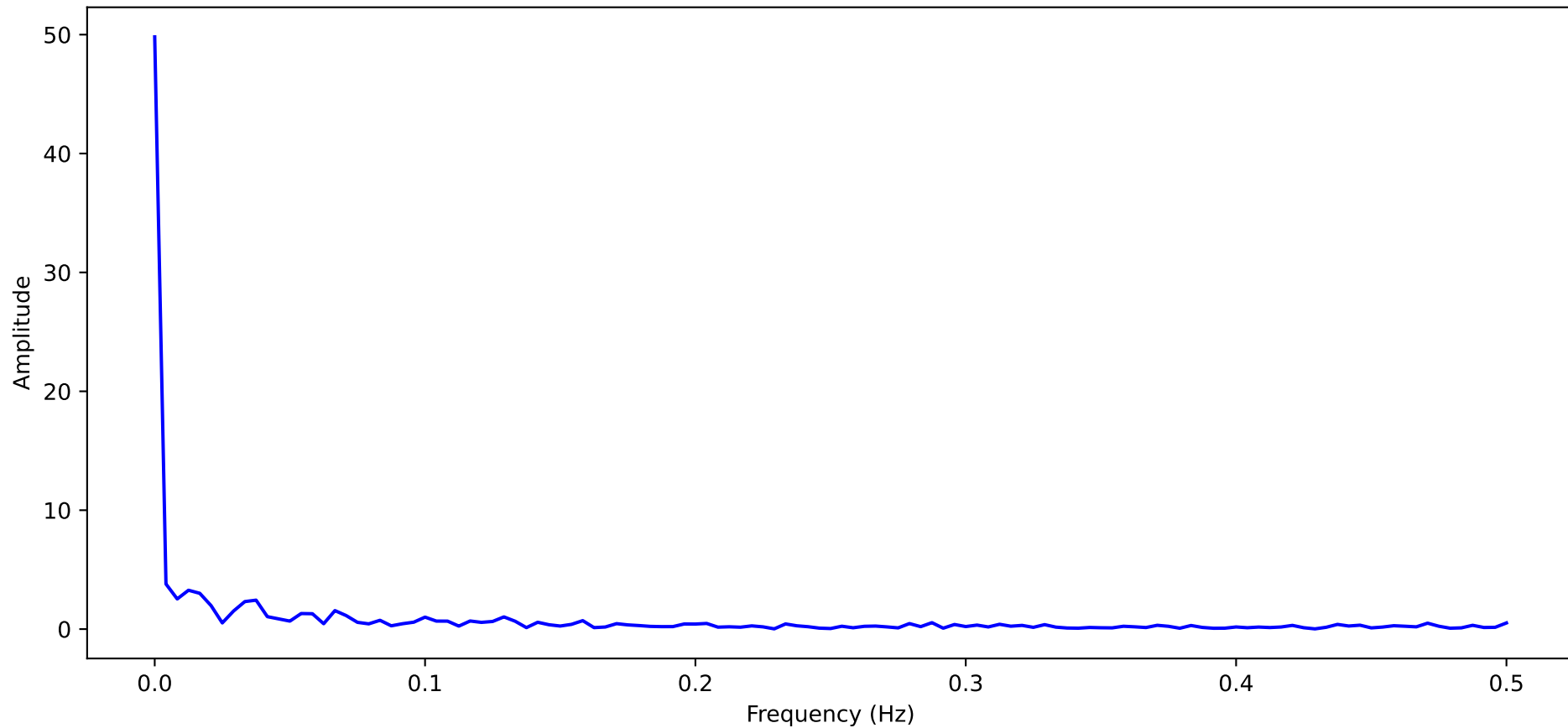


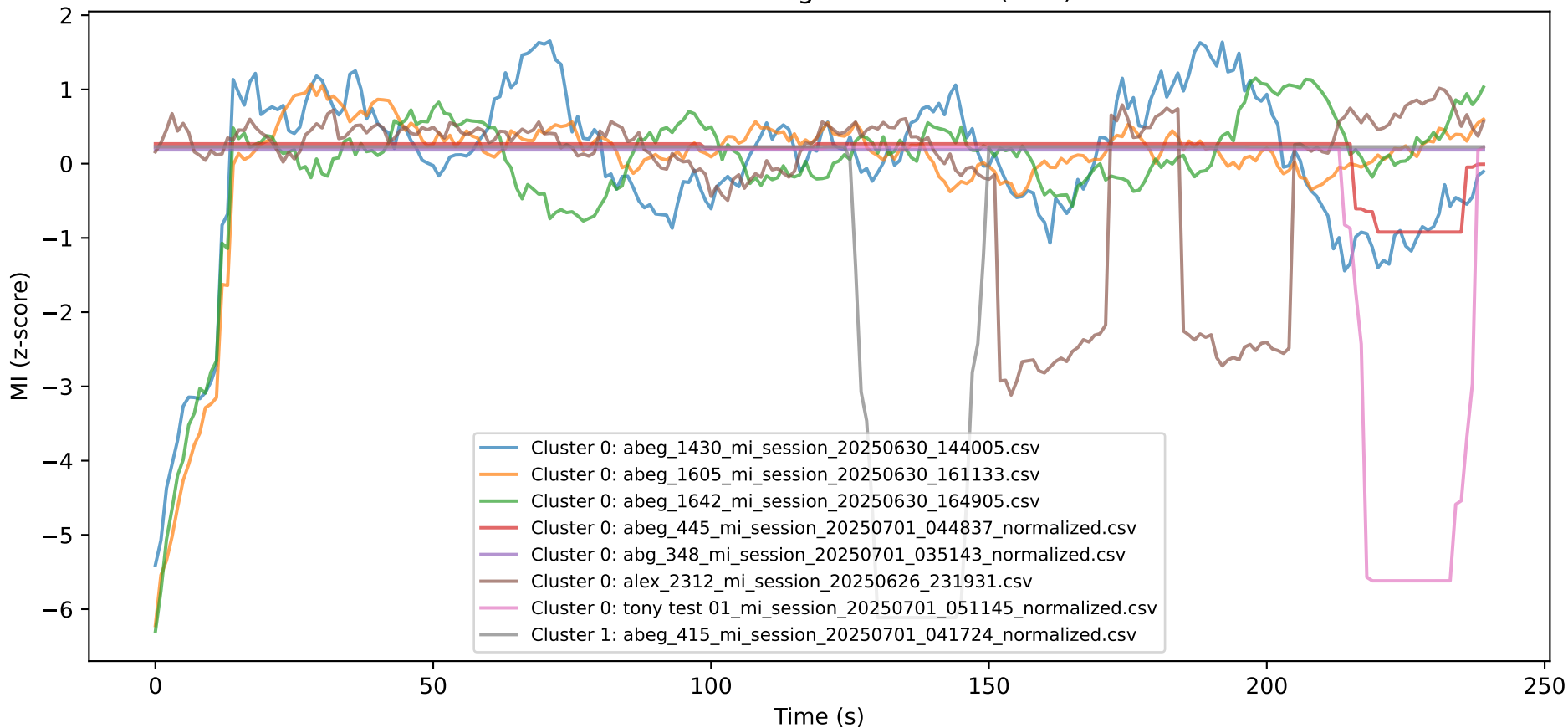
MI Curves (z-score, MA=20) with Median and Peaks



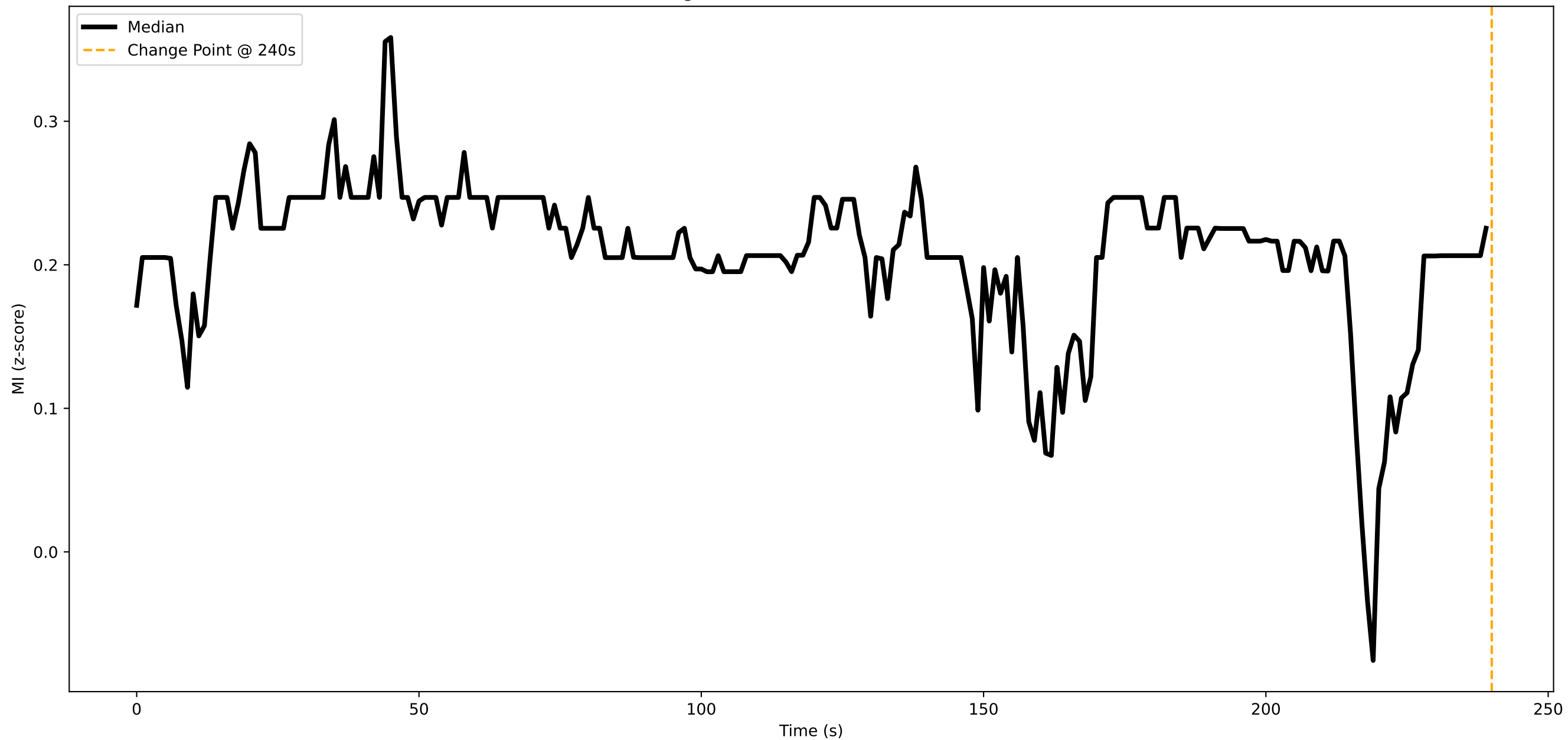
Median MI Frequency Spectrum (FFT)



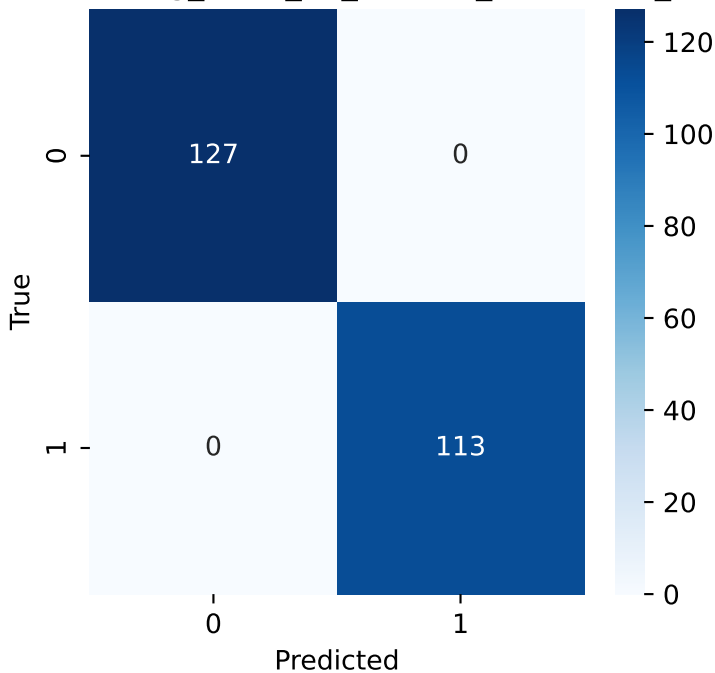
KMeans Clustering of MI Curves (k=2)



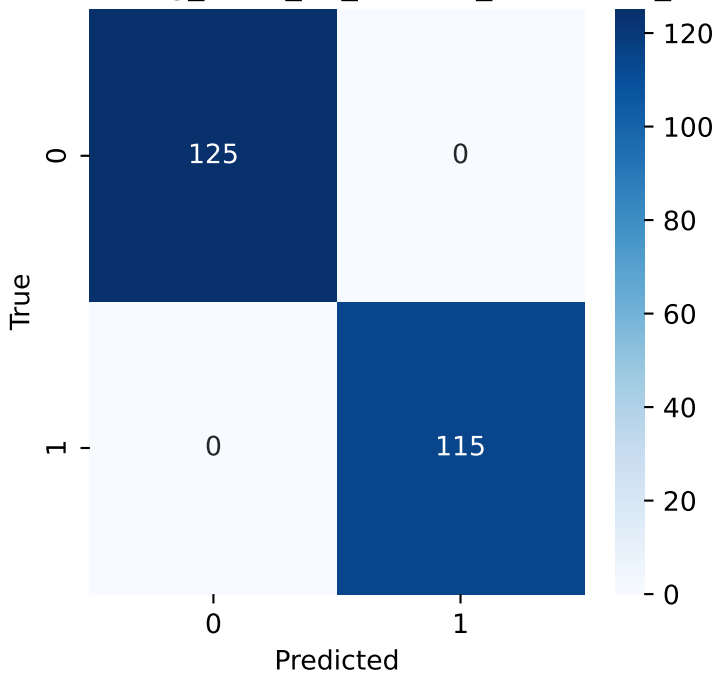
Change Point Detection on Median MI



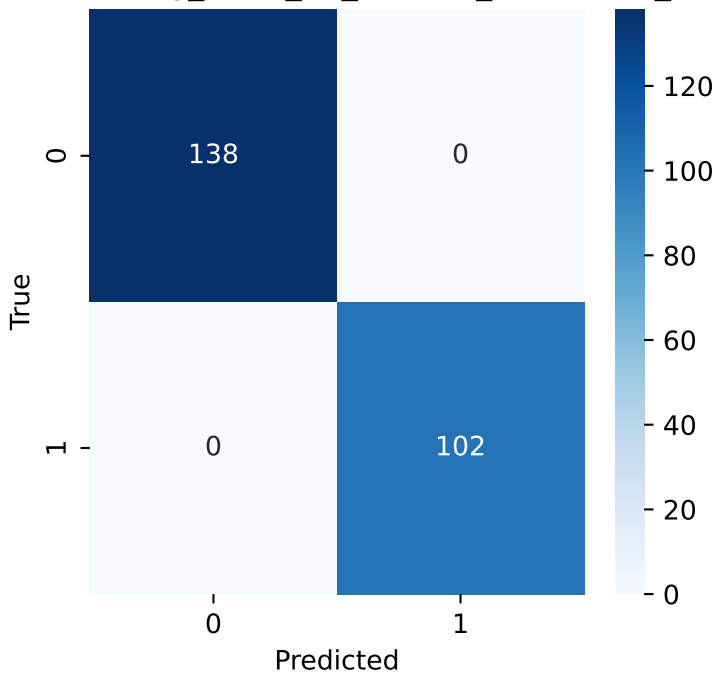
Matrix: abeg_1430_mi_session_20250630_1440



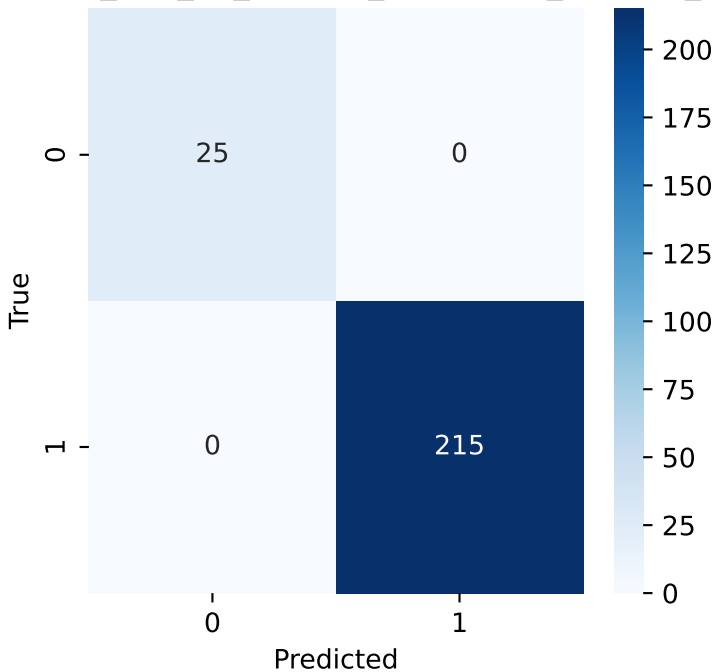
Matrix: abeg_1605_mi_session_20250630_1611



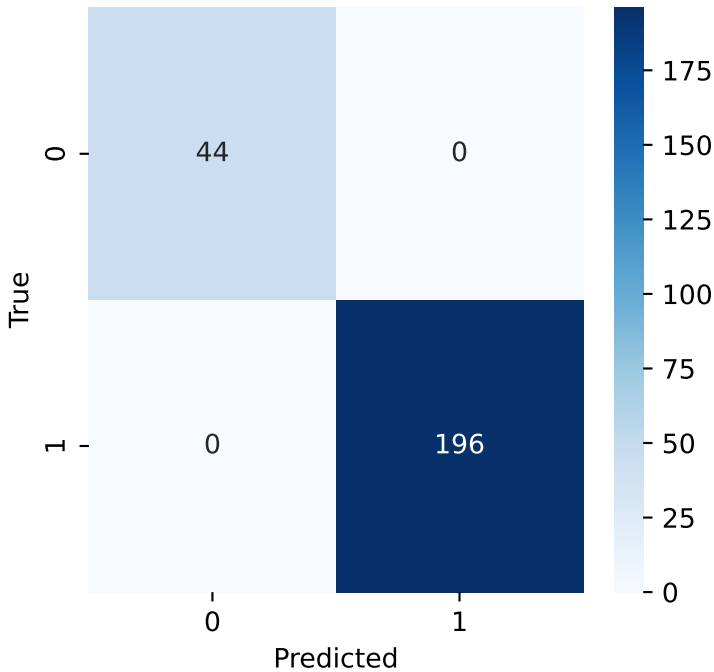
Matrix: abeg_1642_mi_session_20250630_1649



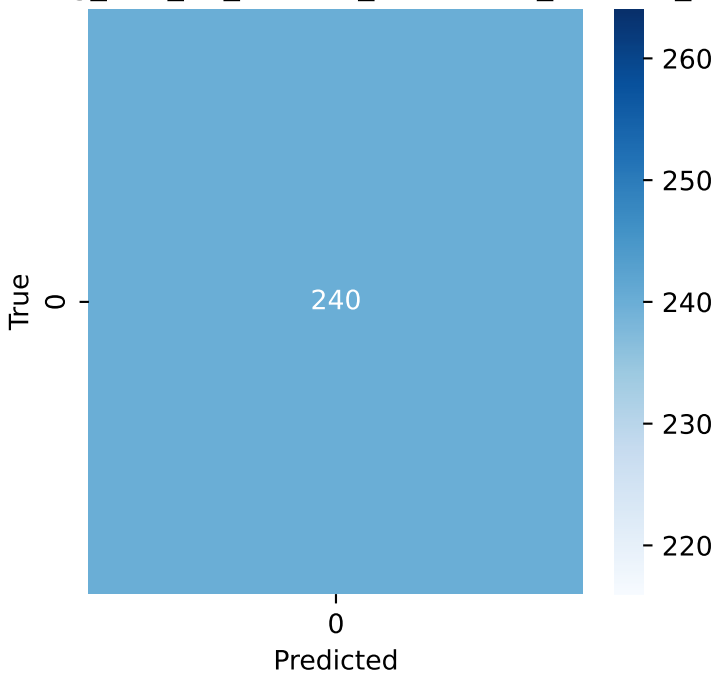
x: abeg_415_mi_session_20250701_041724_no



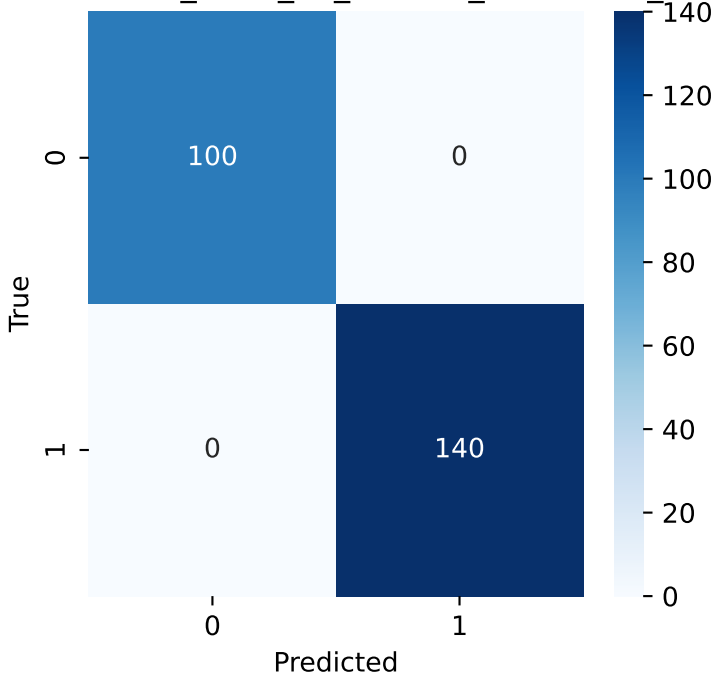
x: abeg_445_mi_session_20250701_044837_no



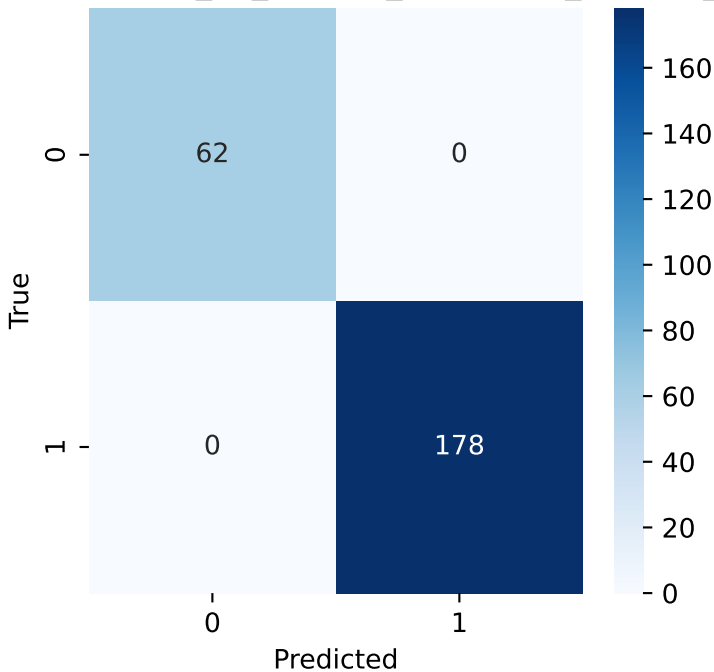
ix: abg_348_mi_session_20250701_035143_nor



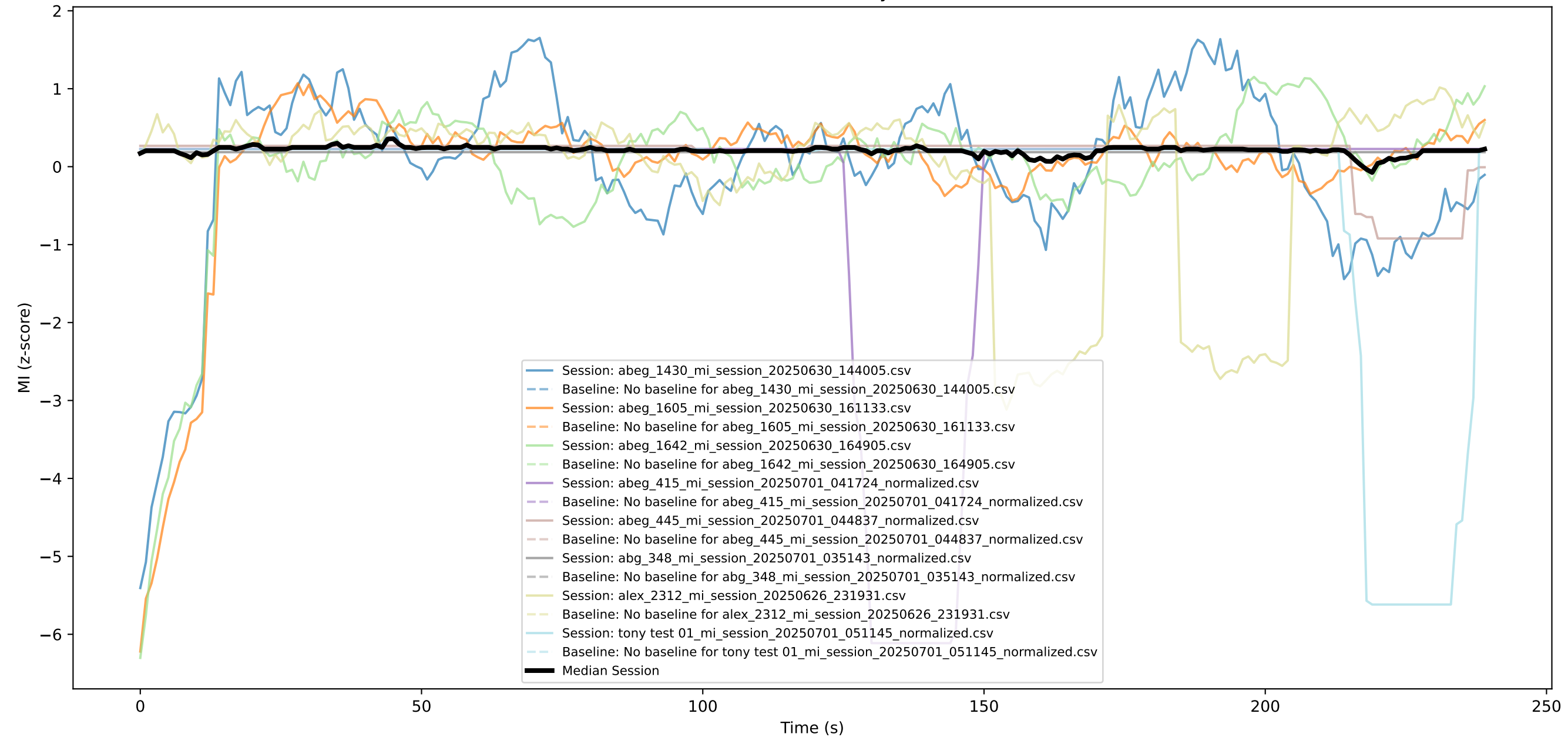
Matrix: alex_2312_mi_session_20250626_2319



ky: tony test 01_mi_session_20250701_051145_no



Session MI vs. Baseline (Eyes Closed)



MI Advanced Report

Generated: 20250717_001353

MI Advanced Report - Statistical Summary

1. Mindfulness Index (MI) Trajectory

- All MI session files with ≥ 240 samples were included.
- Individual and median MI curves were plotted (see Figure 8.1).
- RM-ANOVA across five epochs (0-30s, 31-90s, 91-150s, 151-240s, 241-300s):

Anova

```
=====
      F Value Num DF  Den DF Pr > F
-----
Epoch      nan 4.0000 28.0000      nan
=====
```

- Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1  group2  meandiff p-adj lower upper reject
-----
  0-30s 151-240s  0.2061      nan      nan      nan False
  0-30s 241-300s      nan      nan      nan      nan False
  0-30s  31-90s  0.6349      nan      nan      nan False
  0-30s  91-150s  0.2753      nan      nan      nan False
151-240s 241-300s      nan      nan      nan      nan False
151-240s  31-90s  0.4287      nan      nan      nan False
151-240s  91-150s  0.0692      nan      nan      nan False
241-300s  31-90s      nan      nan      nan      nan False
241-300s  91-150s      nan      nan      nan      nan False
  31-90s  91-150s -0.3596      nan      nan      nan False
=====
```

- Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- Peak analysis (median curve): 0 peaks ($z > 1.5$) at times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- AUC (median): 49.61; mean $\pm$ SD (individual): -10.53 ± 40.79 (see Table 8.3).
- Bootstrap 95% CI for mean MI difference (0-150s vs 151-300s): mean diff = 0.000, CI = [-0.014, 0.014].
- Change point detection (Pelt algorithm): 1 change points at [240] (see Figure 8.5).

3. Clustering & Mixed-Effects

- K-Means clustering (k=2): cluster averages {0: -0.011260313355158206, 1: -0.30154360553767134}, u
- Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

=====

```
Model:          MixedLM Dependent Variable: value
No. Observations: 1920  Method:          REML
No. Groups:      8      Scale:          1.2493
Min. group size: 240    Log-Likelihood: -2953.1571
Max. group size: 240    Converged:       Yes
Mean group size: 240.0
```

```
-----
      Coef. Std.Err.  z  P>|z| [0.025 0.975]
-----
Intercept    0.077   0.077  0.997  0.319 -0.075  0.229
time         -0.001   0.000 -2.835  0.005 -0.002 -0.000
Group Var     0.027   0.016
=====
```

- ANOVA on final epoch by cluster: F=nan, p=nan (see Table 8.5).